KAPU measurements

On PLC manuals of six vendors, Claude Sonnet with KAPU 0.6.0 costs $1.42 where one Markdown file with grep costs $1.85, at about the same accuracy (98 and 99 of 108 answers fully correct). A small local model, qwen3:8b on an 8 GB GPU, gets 1 of 36 questions fully correct alone, 12 with KAPU at 8k context and 21 at 32k. On a knowledge file the model had not seen (PixelLab documentation, 206,899 tokens), an agent using KAPU read 4,104 tokens per request, an agent searching one Markdown file 8,280, and one searching the same text as a PDF 8,207 (median, the same prompt for all three). Correct answers: KAPU 12, Markdown 12, PDF 11 of 12, by three blind judges.

Latest measurements 2026-11-05 · tools 0.6.0, specification v1.5, format 1.3 · Windows 11, release build · every number comes from bench/results/ in the repository

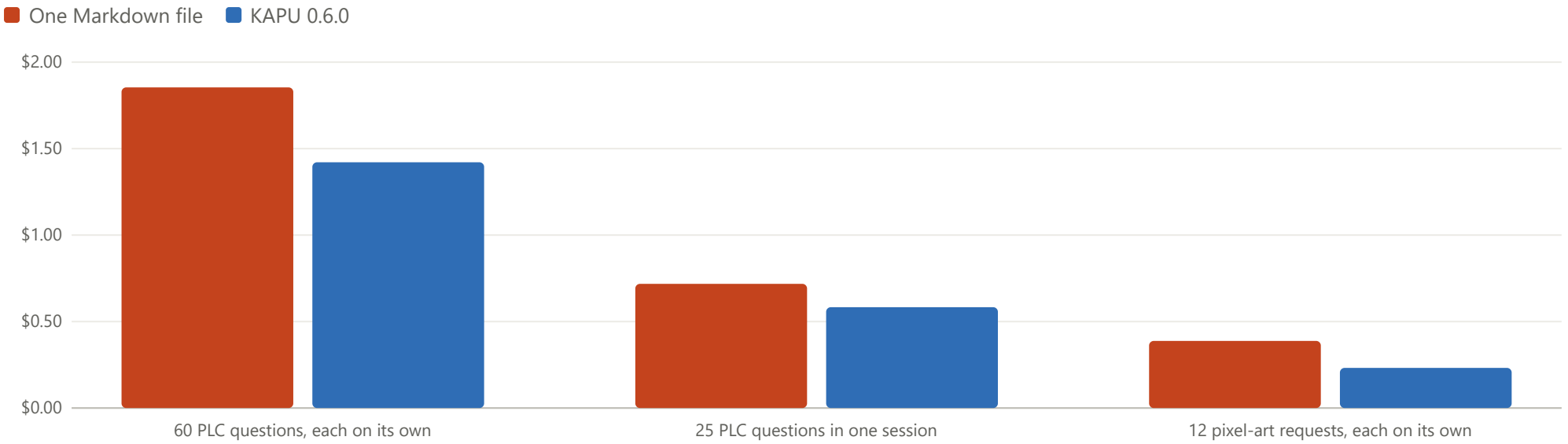
KAPU	One .md file + search	PDF + search	Whole file in the context
4,104	8,280	8,207	206,899
tokens per request (median)	tokens per request (median)	tokens per request (median)	tokens on every request

Against one Markdown file with grep (0.6.0)

Claude Sonnet through claude -p, judged by Claude Haiku, up to 2026-11-05. The public PLC manuals of six vendors as one KAPU file and as one Markdown file with Grep and Read: 36 questions they answer and 24 they do not. Every form starts with one unrecorded call that warms the prompt cache.

Dollars: one Markdown file and KAPU 0.6.0

The same work for both. Accuracy is in the tables below.



60 PLC questions, each on its own, three runs

Dollars for all 60 questions: the mean of three runs and the first run, the fresh price. Fully correct and partly of 108 answers, made up of 72.

Form	Dollars, mean	First run	Fully correct	Partly	Made up
One Markdown file, Grep and Read	$1.85	$2.02	99 / 108	9	5 / 72
KAPU 0.5.1	$2.56	$2.61	99 / 108	9	2 / 72
KAPU 0.6.0	$1.42	$1.50	98 / 108	9	2 / 72

Sessions, one run each

The questions one after the other in one conversation, so what was read stays in the context. One run each cannot tell whether 17 and 19 of 20 differ. The pixel-art documentation is third-party and not published.

Questions	Form	Dollars	Fully correct	Partly	Wrong	Not found
25 PLC (20 answerable)	One Markdown file	$0.72	19 / 20	0	0	1
25 PLC (20 answerable)	KAPU 0.5.1	$1.20	19 / 20	1	0	0
25 PLC (20 answerable)	KAPU 0.6.0	$0.58	17 / 20	2	0	1
12 pixel-art	One Markdown file	$0.52	2 / 12	9	1	0
12 pixel-art	KAPU 0.5.1	$0.42	4 / 12	6	2	0
12 pixel-art	KAPU 0.6.0 before the passage tail	$0.26	4 / 12	8	0	0
12 pixel-art	The whole Markdown file pasted	$1.68	2 / 12	10	0	0

The optional hook and a hard limit, one run each

First-run prices on the 60 PLC questions. The hook hands the kapu_ask passages over with the prompt. The limit of one tool call is not in the code.

Form	Dollars	Of the Markdown file	Fully correct	Made up
One Markdown file, Grep and Read	$2.02	100%	32 / 36	1 / 24
KAPU 0.6.0, tools alone	$1.50	74%	31 / 36	1 / 24
The hook	$1.53	76%	33 / 36	1 / 24
The hook with its stop line	$1.52	75%	35 / 36	1 / 24
The hook and at most one tool call (left out)	$0.90	44%	24 / 36	0 / 24

- KAPU 0.6.0 is cheaper than one Markdown file with grep at about the same accuracy, on single questions and in sessions. Its target of 2/5 of the Markdown file's dollars at the same accuracy was not met.
- What costs money is new text, not rounds: the dollars claude reports fit exactly, on 600 calls, with input 2, cache write 4, cache read 0.1 and output 10 dollars per million tokens.
- The hook saves a round, but the passages have to be written either way, so it is not cheaper than the tools alone. A hard limit of one call is, at the price of accuracy: 8 answerable

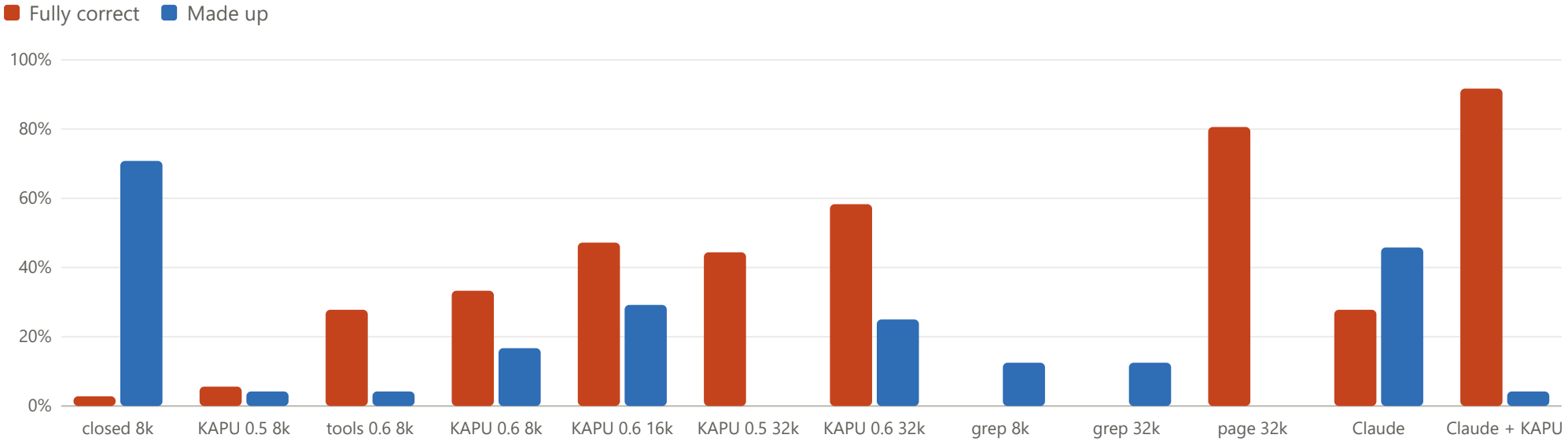
questions became a false not found.

A small local model: qwen3:8b with KAPU

qwen3:8b on an 8 GB GPU (Ollama) through the Vellum client, the way a user runs it, up to 2026-11-05; one run per question, temperature 0, judged by Claude Haiku. In the prefetch forms the client puts a kapu_ask reply into the prompt within its knowledge budget, and the model may call the tools. With 0.6.0 every question has its own kapu-mcp session.

PLC manuals: what the small model gets right

36 questions the manuals answer: the share fully correct. 24 they do not answer: the share made up. Claude Sonnet is the same set with the judge of the Claude runs.



Form	KAPU	Context	Budget	Correct	Partly	Right room read	Made up (absent)	Seconds
Closed book	—	8k	—	3%	22%	—	71%	13
KAPU prefetch + tools	0.5	8k	600	6%	17%	61%	4%	14
KAPU tools	0.6	8k	600	28%	14%	36%	4%	12
KAPU prefetch + tools	0.6	8k	600	33%	14%	47%	17%	12
KAPU prefetch + tools	0.6	16k	4,000	47%	11%	67%	29%	29
KAPU prefetch + tools	0.5	32k	6,000	44%	11%	67%	—	61
KAPU prefetch + tools	0.6	32k	8,000	58%	14%	81%	25%	69
Markdown, grep + line reads	—	8k	—	0%	14%	0%	13%	12
Markdown, grep + line reads	—	32k	—	0%	0%	0%	13%	43
Whole file pasted, cut to fit	—	8k	—	3%	6%	0%	17%	15
The right page in the prompt	—	8k	—	75%	25%	—	—	11
The right page in the prompt	—	32k	—	81%	14%	100%	—	49
Claude Sonnet, closed book	—	—	—	28%	17%	—	46%	—

Form	KAPU	Context	Budget	Correct	Partly	Right room read	Made up (absent)	Seconds
Claude Sonnet with KAPU	0.3	—	—	92%	8%	—	4%	—

Public documentation and PixelLab

Correct / partly / wrong. Rust book, Kubernetes and MDN with Stack Overflow questions, PixelLab with 12 beginner requests (third-party, not published). KAPU 0.6.0; the Rust book at 32k has 29 questions, because one long question got no reply within the client's three-minute limit.

Set	Closed book	KAPU tools, 8k	KAPU prefetch, 8k	KAPU prefetch, 32k	Markdown, grep, 8k
Rust book (30)	17% / 70% / 13%	13% / 83% / 3%	13% / 87% / 0%	17% / 79% / 3%	10% / 60% / 30%
Kubernetes (30)	10% / 70% / 17%	—	13% / 80% / 3%	10% / 80% / 7%	7% / 37% / 20%
MDN (30)	30% / 63% / 3%	—	20% / 63% / 3%	30% / 57% / 3%	33% / 30% / 10%
PixelLab (12)	0% / 25% / 67%	8% / 42% / 8%	0% / 50% / 42%	0% / 75% / 25%	0% / 25% / 0%

- On manuals it cannot know, 0.6.0 moves the small model: 1 of 36 fully correct alone, 0 with grep over the same text as Markdown, 2 with KAPU 0.5 and 12 with KAPU 0.6 at 8k context, on the same card (sign test: p = 0.003 against the model alone, p = 0.013 against 0.5). At 32k 21 of 36, where 0.5 had 16; that last step is not a measured difference on this few questions.
- It makes up far less than alone, but more than 0.5: of the 24 questions the manuals do not answer, 17 alone, 1 with 0.5 at 8k, 4 with 0.6 at 8k, 7 at 16k and 6 at 32k. The tools alone at 8k make up 1, with 10 fully correct.
- Where the model half knows the subject from training, KAPU changes little: fully correct on the Rust book 17% alone and 13-17% with KAPU, Kubernetes 10% and 10-13%, MDN 30% and 20-30%. The wrong answers do fall on the Rust book and on Kubernetes. On PixelLab 1 of 12 at best.
- The gap is still reading: with the right page in the prompt the same model gets 27 to 29 of 36, and Claude Sonnet with KAPU 33.
- The price on an 8 GB card is time: 12 seconds a question at 8k, 29 at 16k, 69 at 32k on the manuals, 157 on the Rust book, where one long question got no reply within the client's three-minute limit.

Plan v3: a contextual model in plain Rust (0.5.0)

Measured 2026-10-28 with tools 0.5.0, specification v1.3, format 1.3, on the seven question sets and the dev/test split of plan v2 (577 test questions), the test part reported. The static model of 0.3 limited the hybrid search, so plan v3 measured other embedding models and built the winner, multilingual-e5-small, into KAPU: its own Rust code runs the model with exact integer products, so a vector has the same bits on every machine, and the file stores every vector at 4 bits a component.

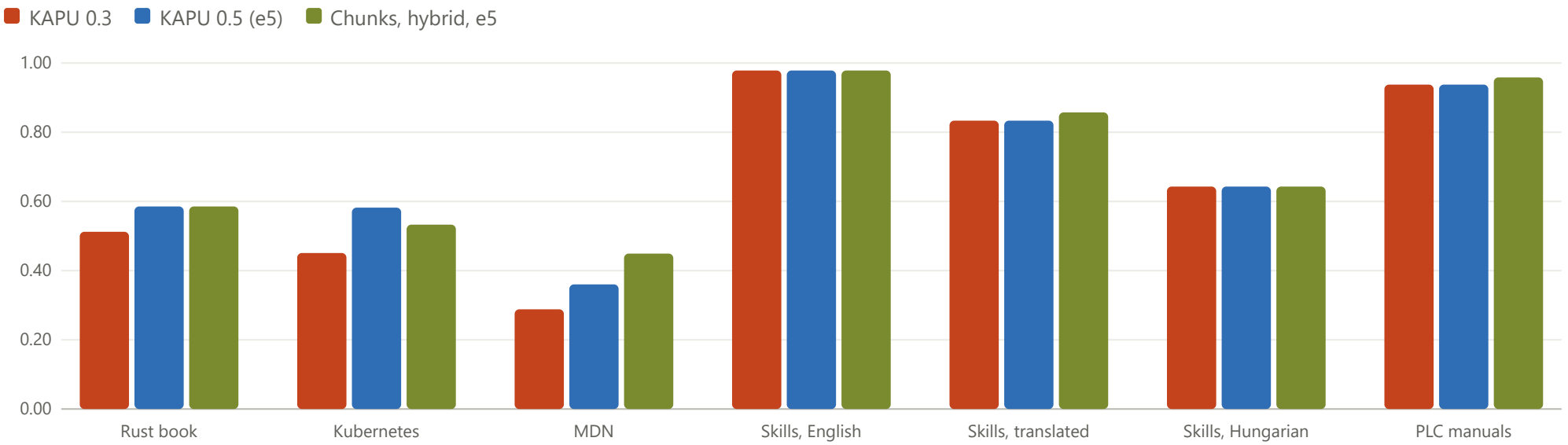
The model swap in KAPU's own search

Difference to the static model of 0.3 with its default ranking, test part, all sets pooled, 95% paired bootstrap interval. Worst set: the largest loss of one set in either metric; p50 ratio: the slowest set's median search time against 0.3. The swap does not pass the M0 rule of 0.3.0 (worst set at most -0.01, at most 1.5 times slower); it was accepted for 0.5.0 with these costs.

Model	Bits	Ranking	Δ room recall@10	Δ room MRR@10	Worst set	p50 ratio
multilingual-e5-small	8	Default ranking	+0.062 [+0.035, +0.090]	+0.027 [+0.005, +0.049]	−0.065	14.9×
multilingual-e5-small	4	Default ranking	+0.062 [+0.035, +0.090]	+0.028 [+0.006, +0.052]	−0.056	14.9×
multilingual-e5-small	4	Passage BM25	+0.069 [+0.038, +0.101]	+0.038 [+0.014, +0.062]	−0.070	14.6×
multilingual-e5-small	4	Passage BM25 and room fusion	+0.075 [+0.045, +0.106]	+0.038 [+0.014, +0.063]	−0.085	14.5×

Per set: 0.3, 0.5 and the chunked hybrid with the same model

Room-level recall@10 on the test part, default ranking, 4 bits. Search time: the median of kapu bench, with the query's vector built for AVX-512, and with the baseline (SSE2) build of the release.



Question set	Questions	recall@10, 0.3	recall@10, 0.5	MRR@10, 0.3	MRR@10, 0.5	Search, 0.3	Search, 0.5	0.5, SSE2
Rust book	41	0.512	0.585	0.252	0.263	0.8 ms	12.6 ms	35.0 ms
Kubernetes	122	0.451	0.582	0.253	0.271	27.1 ms	25.7 ms	48.4 ms
MDN	236	0.288	0.360	0.135	0.195	408.5 ms	155.9 ms	320.4 ms
Skills, English	46	0.978	0.978	0.853	0.868	10.1 ms	17.2 ms	43.1 ms
Skills, translated	42	0.833	0.833	0.759	0.770	12.7 ms	17.6 ms	—
Skills, Hungarian	42	0.643	0.643	0.492	0.436	10.7 ms	18.9 ms	—
PLC manuals	48	0.938	0.938	0.734	0.751	29.7 ms	30.1 ms	—

What it costs: file size and build time

The first build with the model on 12 threads (AVX-512 build). A rebuild takes the computed windows from a log in the cache folder and computes only new text.

File	Size, 0.3	Size, 0.5	Growth	With 8 bits	First build
Rust book	2.4 MB	2.6 MB	+9%	+20%	71 s
Kubernetes	28.1 MB	31.4 MB	+12%	+25%	922 s
MDN	139.0 MB	160.7 MB	+16%	+34%	2,986 s
Skills, English	5.9 MB	6.5 MB	+10%	+21%	124 s
PLC manuals	12.3 MB	13.4 MB	+9%	+20%	321 s

How to store a vector (simulated on the chunks)

KAPU BM25 fused with the chunk vectors, difference to float vectors. Centred: the mean of all vectors taken off first; residual: the code of the vector minus the mean, scored with the original cosine. The sign bits of 0.3 break down with this model, 4 bits cost nothing measurable in KAPU itself.

Storage	Bytes per window	Δ room recall@10	Δ room MRR@10
float, centred	1,536	−0.021 [−0.042, +0.000]	−0.002 [−0.015, +0.011]
sign bits, then 8 bits	432	**−0.121 [−0.154, −0.088]**	**−0.066 [−0.089, −0.044]**
8 bits	384	+0.002 [+0.000, +0.005]	**+0.003 [+0.001, +0.006]**
sign bits, then 4 bits	240	**−0.127 [−0.159, −0.095]**	**−0.074 [−0.096, −0.051]**
centred sign bits, then 4 bits	240	−0.012 [−0.033, +0.009]	−0.001 [−0.014, +0.011]
4 bits, residual	194	−0.009 [−0.023, +0.005]	−0.002 [−0.010, +0.006]
4 bits	192	−0.012 [−0.028, +0.003]	−0.001 [−0.010, +0.007]
4 bits, centred	192	−0.017 [−0.038, +0.003]	−0.004 [−0.016, +0.009]
3 steps, centred	144	−0.019 [−0.040, +0.002]	+0.002 [−0.011, +0.015]
-1/0/+1, residual	98	**−0.035 [−0.055, −0.014]**	**−0.024 [−0.041, −0.007]**
-1/0/+1, centred	96	−0.021 [−0.043, +0.002]	−0.005 [−0.018, +0.009]
sign bits, residual	50	**−0.080 [−0.109, −0.052]**	**−0.063 [−0.085, −0.041]**
sign bits (as in 0.3)	48	**−0.203 [−0.239, −0.168]**	**−0.178 [−0.208, −0.149]**
sign bits, centred	48	**−0.033 [−0.055, −0.010]**	−0.010 [−0.025, +0.005]

The candidates on the same chunks (V1)

Three models on the 512-token chunks, in Python, difference to the static model of 0.3: the vectors alone, the usual chunked hybrid, and KAPU's BM25 ranking fused with the model's vectors at room level.

Model	Ranking	Δ room recall@10	Δ room MRR@10
potion-retrieval-32M	Vectors alone	+0.021 [−0.017, +0.059]	+0.016 [−0.010, +0.042]
potion-retrieval-32M	Chunked hybrid	−0.007 [−0.033, +0.019]	+0.002 [−0.019, +0.023]
potion-retrieval-32M	KAPU BM25 + vectors	+0.003 [−0.026, +0.033]	+0.009 [−0.013, +0.032]
potion-retrieval-32M	KAPU passage BM25 + vectors	+0.009 [−0.021, +0.038]	+0.022 [−0.000, +0.044]
multilingual-e5-small-qint8	Vectors alone	+0.123 [+0.083, +0.163]	+0.072 [+0.042, +0.102]
multilingual-e5-small-qint8	Chunked hybrid	+0.064 [+0.036, +0.092]	+0.039 [+0.016, +0.062]
multilingual-e5-small-qint8	KAPU BM25 + vectors	+0.071 [+0.040, +0.102]	+0.039 [+0.016, +0.063]
multilingual-e5-small-qint8	KAPU passage BM25 + vectors	+0.075 [+0.045, +0.104]	+0.053 [+0.030, +0.076]

- The model is the lever: with multilingual-e5-small, room recall@10 improves by +0.062 with the default ranking and by +0.075 with passage BM25 and room fusion, both intervals above zero. Kubernetes 0.451 to 0.582, MDN 0.288 to 0.360.
- It costs: Hungarian questions on English skills lose MRR (0.492 to 0.436), a small file is searched 12.6 ms instead of 0.8 ms (the query's vector; 35.0 ms with the baseline build), and the files grow by 9-16%. A large file is faster (408.5 ms to 155.9 ms), because nothing is rescored when it is read.
- With the same model, the usual chunked hybrid is 0.585 on the Rust book, 0.533 on Kubernetes and 0.449 on MDN: level with KAPU 0.5, behind it and ahead of it. KAPU's advantage stays the section hit with its token price, in one file with no server.
- The encoder is KAPU's own Rust code, with no new dependency and nothing to download at run time: integer products, fixed-order float sums and its own exp and erf, so the SSE2, AVX2 and AVX-512 builds write the same bytes. It agrees with ONNX Runtime to a cosine of 0.998-1.000; one 512-token window takes 0.45 s on one thread (AVX-512), 1.3 s with the baseline build.

Plan v2, phases 1 to 3: an ablation framework, one fix that holds, honest baselines

Measured 2026-10-27 with tools 0.3.0. Every ranking change of plan v2 is a switch that is off by default; off, the ranking is bit for bit that of 0.2.1. The questions of each set are split once by a

hash of the question into a dev part (40%, 356 questions in all) and a test part (60%, 577): the variant of each fix is chosen on the dev part, and only the test part is reported. A fix is kept when on the test part room recall@10 or room MRR@10 improves with a 95% paired bootstrap interval above zero (all sets pooled), no set loses more than 0.01 and search does not slow down by more than 1.5x.

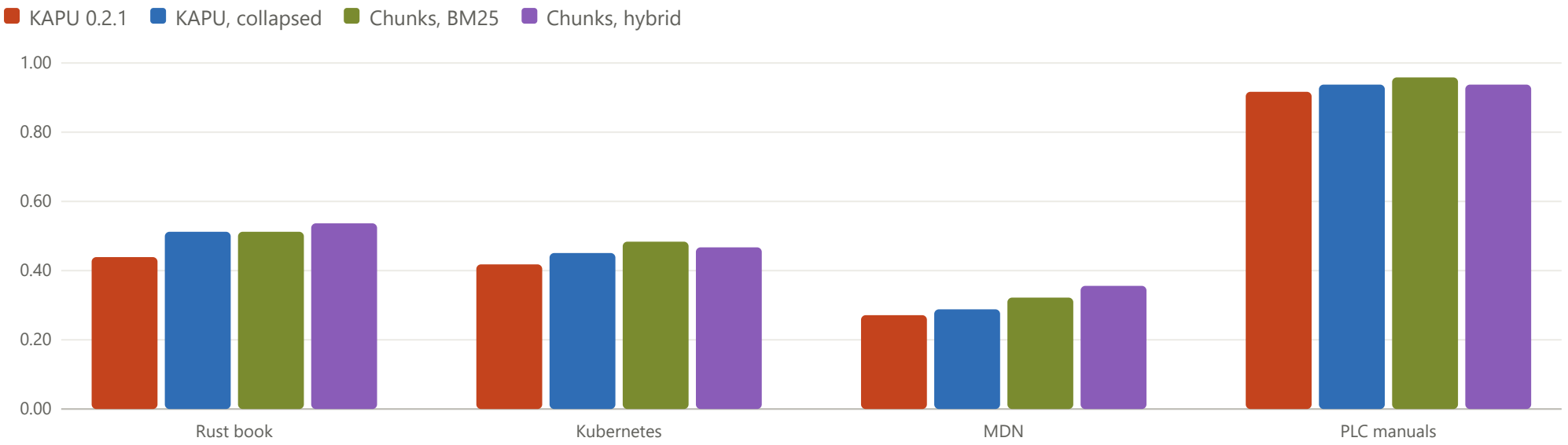
Which fixes pass

Difference to the base ranking on the test part, all sets pooled, with the 95% bootstrap interval. Green: the interval is above zero; red: below. Worst set: the largest loss of one set in either metric.

Fix	Variant chosen on dev	Δ room recall@10	Δ room MRR@10	Worst set	Kept
Stemming: the word only, English or Hungarian stem (M0 axis)	stem-en	+0.002 [−0.012, +0.016]	−0.005 [−0.014, +0.004]	−0.021	no
J2a: separate average length for rooms and sections	avgdl-split	+0.021 [+0.005, +0.036]	+0.016 [+0.003, +0.028]	−0.022	no
J2b: a section scores a share of its room	parent-0.1	−0.004 [−0.019, +0.012]	−0.009 [−0.018, −0.000]	−0.030	no
J1: one hit per room, the other matching sections named	collapse-1	+0.022 [+0.010, +0.035]	+0.011 [+0.008, +0.014]	+0.000	yes
J4a: the first 50 BM25 hits always get a vector score	rescore-bm25-50	+0.002 [−0.005, +0.010]	+0.001 [−0.002, +0.005]	−0.021	no
J10: sections under 40 tokens only through their room	min-door-40	−0.002 [−0.009, +0.004]	+0.001 [−0.001, +0.004]	−0.009	no
J4c: weights by query (code words, unknown words)	adaptive-1.5	+0.002 [−0.005, +0.009]	+0.000 [−0.003, +0.003]	−0.021	no
J4b: weighted RRF (BM25 1-3, vectors 0.5-1, k 20 or 60)	rrf-b1-v0.7-k60	+0.009 [−0.005, +0.024]	+0.006 [+0.001, +0.012]	−0.048	no
J4d: min-max scores instead of ranks	convex-b1-v1	+0.002 [−0.017, +0.021]	+0.004 [−0.008, +0.016]	−0.024	no

Retrieval against the chunk baselines (test part)

Room-level: a hit is any door of the expected page. The chunk baselines cut every page into 512-token chunks and count a page once (deduplicated), which is the fair comparison with KAPU's room level. Collapsed: one hit per room (J1), the new default.



Question set	Questions	Ranking	recall@1	Recall@10	recall@100	MRR@10	nDCG@10
Rust book	41	KAPU 0.2.1	0.146	0.439	0.902	0.220	0.271
Rust book	41	KAPU, collapsed	0.146	0.512	0.902	0.252	0.314
Rust book	41	Chunks, BM25	0.146	0.512	0.902	0.255	0.315
Rust book	41	Chunks, vectors	0.122	0.439	0.854	0.223	0.274
Rust book	41	Chunks, hybrid	0.146	0.537	0.829	0.249	0.317
Kubernetes	122	KAPU 0.2.1	0.139	0.418	0.713	0.238	0.282
Kubernetes	122	KAPU, collapsed	0.139	0.451	0.713	0.253	0.302
Kubernetes	122	Chunks, BM25	0.164	0.484	0.795	0.242	0.299
Kubernetes	122	Chunks, vectors	0.066	0.328	0.639	0.145	0.189
Kubernetes	122	Chunks, hybrid	0.164	0.467	0.738	0.250	0.301
MDN	236	KAPU 0.2.1	0.076	0.271	0.525	0.127	0.161
MDN	236	KAPU, collapsed	0.076	0.288	0.525	0.135	0.171

Question set	Questions	Ranking	recall@1	Recall@10	recall@100	MRR@10	nDCG@10
MDN	236	Chunks, BM25	0.089	0.322	0.602	0.156	0.196
MDN	236	Chunks, vectors	0.064	0.203	0.517	0.095	0.120
MDN	236	Chunks, hybrid	0.064	0.356	0.631	0.145	0.195
Skills, English	46	KAPU 0.2.1	0.783	0.978	1.000	0.846	0.878
Skills, English	46	KAPU, collapsed	0.783	0.978	1.000	0.853	0.884
Skills, translated	42	KAPU 0.2.1	0.714	0.833	0.833	0.756	0.775
Skills, translated	42	KAPU, collapsed	0.714	0.833	0.833	0.759	0.777
Skills, Hungarian	42	KAPU 0.2.1	0.405	0.619	0.762	0.487	0.520
Skills, Hungarian	42	KAPU, collapsed	0.405	0.643	0.762	0.492	0.529
PLC manuals	48	KAPU 0.2.1	0.625	0.917	0.979	0.720	0.767
PLC manuals	48	KAPU, collapsed	0.625	0.938	0.979	0.734	0.783
PLC manuals	48	Chunks, BM25	0.625	0.958	0.979	0.741	0.794
PLC manuals	48	Chunks, vectors	0.458	0.792	0.896	0.571	0.625
PLC manuals	48	Chunks, hybrid	0.583	0.938	0.979	0.686	0.746

Why a question is missed (M3)

Every test question whose page is not in the hybrid top 10, by the first class that fits: duplication (the page would be in the top 10 with one hit per room), fusion (one ranking had it in its top 10), rescoring (a good BM25 hit without a vector score), ranking (in the top 100), vocabulary (nowhere). Merge: the page is in the top 10 but the expected section is not.

Question set	Ranking	Missed	Duplication	Fusion	Rescoring	Ranking	Vocabulary	Merge
Rust book	KAPU 0.2.1	23	3	5	1	10	4	1
Rust book	KAPU, collapsed	20	0	5	1	10	4	8
Kubernetes	KAPU 0.2.1	71	4	11	6	15	35	9
Kubernetes	KAPU, collapsed	67	0	11	6	15	35	26
MDN	KAPU 0.2.1	172	4	17	15	24	112	8
MDN	KAPU, collapsed	168	0	17	15	24	112	23
Skills, English	KAPU 0.2.1	1	0	0	0	1	0	2
Skills, English	KAPU, collapsed	1	0	0	0	1	0	15
Skills, translated	KAPU 0.2.1	7	0	0	0	0	7	1
Skills, translated	KAPU, collapsed	7	0	0	0	0	7	7
Skills, Hungarian	KAPU 0.2.1	16	1	1	0	4	10	2
Skills, Hungarian	KAPU, collapsed	15	0	1	0	4	10	7
PLC manuals	KAPU 0.2.1	4	1	1	0	1	1	1
PLC manuals	KAPU, collapsed	3	0	1	0	1	1	21

kapu_ask: what one call opens (J12)

Answered: a frame of the expected page is among the opened ones. Spread: the start of the best door of each of the first 5 rooms, the 2000-token budget shared. Confidence: the best hit alone when it is clearly ahead, otherwise spread, and the reply says confidence: high or low. Tokens without the page: the tokens of the replies that missed it, per question. Search hits: how many fit in the 600-token search reply.

Question set	Mode	Answered	Reply tokens	Tokens without the page	Search hits
Rust book	Best hit (collapsed)	15%	1,662	1,336	7.2
Rust book	Best hit (0.2.1)	15%	1,653	1,315	10.0
Rust book	Spread over 5 rooms	41%	1,224	710	7.2
Rust book	Excerpts only, nothing opened	0%	215	211	7.2
Rust book	Confidence, ratio 1.1	37%	1,239	835	7.2
Rust book	Confidence, ratio 1.3	41%	1,226	711	7.2
Rust book	Confidence, ratio 1.6	41%	1,226	713	7.2
Rust book	Best hit, two-line excerpts (J11)	15%	1,664	1,336	5.9
Kubernetes	Best hit (collapsed)	17%	1,824	1,404	8.6
Kubernetes	Best hit (0.2.1)	17%	1,800	1,391	9.9
Kubernetes	Spread over 5 rooms	39%	1,130	678	8.6
Kubernetes	Confidence, ratio 1.3	39%	1,155	721	8.6
Kubernetes	Best hit, two-line excerpts (J11)	17%	1,823	1,406	7.0
MDN	Best hit (collapsed)	12%	1,678	1,344	9.5
MDN	Best hit (0.2.1)	11%	1,614	1,331	10.0
MDN	Spread over 5 rooms	22%	1,124	854	9.5
MDN	Confidence, ratio 1.3	21%	1,140	892	9.5
MDN	Best hit, two-line excerpts (J11)	12%	1,682	1,344	7.8
Skills, English	Best hit (collapsed)	83%	1,827	301	5.3
Skills, English	Best hit (0.2.1)	83%	1,788	295	9.8
Skills, English	Spread over 5 rooms	96%	1,145	51	5.3
Skills, English	Excerpts only, nothing opened	0%	209	210	5.3
Skills, English	Confidence, ratio 1.1	93%	1,186	85	5.3
Skills, English	Confidence, ratio 1.3	96%	1,151	52	5.3
Skills, English	Confidence, ratio 1.6	96%	1,148	51	5.3

Question set	Mode	Answered	Reply tokens	Tokens without the page	Search hits
Skills, English	Best hit, two-line excerpts (J11)	83%	1,828	300	4.7
Skills, translated	Best hit (collapsed)	76%	1,794	372	5.3
Skills, translated	Best hit (0.2.1)	76%	1,684	384	9.6
Skills, translated	Spread over 5 rooms	81%	1,144	210	5.3
Skills, translated	Excerpts only, nothing opened	0%	217	217	5.3
Skills, translated	Confidence, ratio 1.1	79%	1,193	249	5.3
Skills, translated	Confidence, ratio 1.3	81%	1,148	210	5.3
Skills, translated	Confidence, ratio 1.6	81%	1,148	211	5.3
Skills, translated	Best hit, two-line excerpts (J11)	76%	1,796	372	4.6

Do the topic groups match the documentation's folders? (N13)

Second folder level of the source (Kubernetes concepts/workloads, MDN web/css). ARI: 0 is random, 1 the same grouping; NMI in brackets for random groups of the same sizes. k-means: the same number of groups from the room vectors. sim and term: Leiden on only that edge type. Largest group: rooms / tokens.

Question set	Rooms	Groups / folders	ARI	NMI (random)	k-means ARI	sim only	term only	Largest group
Kubernetes	1,722	104 / 73	0.159	0.416 (0.150)	0.137	0.155	−0.005	166 / 1,509,243
MDN	14,719	99 / 50	0.040	0.390 (0.029)	0.032	0.041	−0.001	733 / 1,296,318

- One fix holds and is the default in 0.3.0: one hit per room, with the room's other matching sections named under it. On the test part, all seven sets pooled: room recall@10 +0.022 [95% +0.010, +0.035], MRR@10 +0.011 [+0.008, +0.014], no set worse, search time unchanged (Kubernetes p50 27.3 to 27.9 ms, MDN 313.5 to 313.8 ms).
- Three changes come close but lose one question on a small set, more than the rule allows: separate average lengths for rooms and sections (+0.021 recall@10), BM25 weighted 1.5 with k 20 on top of the collapse (+0.028), and the same with a query-dependent weight (+0.032). They stay switches, off. The graph step, other fusions, stemming variants and title weights give nothing measurable.
- Honest baselines: page-deduplicated chunk BM25 is still ahead of KAPU on Kubernetes (0.484 against 0.451) and MDN (0.322 against 0.288), the chunk hybrid on every set but the PLC manuals. Until 0.2.1 the README compared with chunks counted per chunk. MDN's misses are mostly vocabulary: 112 of 168 pages are in no top 100.

- kapu_ask now opens the start of the first 5 rooms in one shared budget: the expected room is in the reply for 42% instead of 15% (Rust book), 39% instead of 17% (Kubernetes), 22% instead of 12% (MDN) and 96% instead of 83% (skills), in about 1,130 instead of 1,800 tokens. The planned confidence signal is not calibrated (where it says high, 11% on Kubernetes and MDN), so it is not shown.
- On PLC manuals a model does not know (six vendors, 623 pages), Claude Sonnet answered 92% correctly with the KAPU tools, 97% with the right page in the prompt and 28% alone. For 24 questions the manuals do not answer, the KAPU agent said so 23 times; alone the model made an answer up 11 times. One run per question; the question writer saw the pages.
- The graph helps navigation, not ranking: of 27 later opens in the logged live runs, 5 (19%) came from a neighbour list (link 2 of 64 shown, sim 3 of 64, term 0 of 221), at 70 tokens per opened door. The topic groups follow the documentation's folders at the level of k-means on the room vectors, and the sim edges carry that structure.

File format 1.2: vectors in the file, replies within a budget

Measured 2026-10-26 with tools 0.2.0, specification v1.2, file format 1.2, on the same corpora, questions and machine as format 1.1, and on a fourth corpus: agent skills and frontend notes with 77 English and 60 Hungarian questions. The search vectors are now in the file, one sign bit per component of every window vector. The sign bits pick the 200 best candidates and the model's float vectors order them, so the first search on another machine needs no cache. Every MCP reply starts with its own price and stays within a max_tokens budget, and kapu_ask searches and opens in one call.

Requirements of the second protocol

The measurable requirements of the 2026-10-26 protocol and their acceptance limits. Tokens: o200k_base.

ID	Requirement	Target	Measured	Met
K-01	Tool descriptions and input schemas, all six tools	≤ 800	797	yes
K-02	First map reply on a file of 3,000 rooms	≤ 1,000	306	yes
K-05	Search reply, 10 hits with excerpts (largest on any corpus)	≤ 600	586	yes
K-07	Questions answered by one kapu_ask call (skills corpus)	≥ 70%	83.1%	yes
K-11	File growth from the #VEC section (largest corpus, MDN)	≤ 3%	3.3%	no
K-12	Hungarian questions on English skills, translated: hybrid recall@10	≥ 0.75	0.78	yes
K-13	The same without translation (needs another embedding model)	≥ 0.60	0.55	no
K-22	Opening a door on a file of 3,000 rooms (median)	< 20 ms	7.6 ms	yes

Four corpora in format 1.2

File size before and after the vector section. Times and reply sizes are medians over MCP, except the largest search reply; tokens are what the agent receives.

Corpus	File, v1.1 (MiB)	File, v1.2 (MiB)	Growth	Search (ms)	Open (ms)	First map (tokens)	Next map (tokens)	Search (tokens)	Largest search (tokens)	kapu_ask (tokens)	Answered by one kapu_ask
Rust book	2.21	2.24	1.7%	32	3.6	328	159	527	580	1,657	21%
Kubernetes	26.22	26.78	2.2%	50	7.9	967	983	528	586	1,795	20%
MDN	128.34	132.52	3.3%	282	48	969	980	500	583	1,664	12%
Skills and notes	5.56	5.67	2.0%	26	2.9	600	224	547	585	1,786	83%

Retrieval: room-level recall@10, hybrid ranking

Sign bits alone lost on MDN, on the skills and most of all with Hungarian questions on English text. With the float rescoring of the best candidates they are at the float level or above everywhere; the price is search time.

Question set	Questions	v1.1, float vectors	v1.2, sign bits alone	v1.2, sign bits + rescoring
Rust book (Stack Overflow)	68	0.485	0.500	0.485
Kubernetes (Stack Overflow)	201	0.458	0.458	0.458
MDN (Stack Overflow)	395	0.271	0.246	**0.278**
Skills, English questions	77	0.948	0.922	**0.961**
Skills, Hungarian questions	60	0.633	0.583	0.633
Skills, the Hungarian questions translated	60	—	0.833	0.833

Not measured yet

- Skill finding with 250 skills (the test machine has 91), and with a prompt that does not mention skills, where the agent has to decide on its own.
- Clients other than Claude Code: the configurations for the Codex CLI and VS Code are in the README, but they were not tried, and no measurement with them is planned.

Finding skills: native Claude Code skills, one KAPU file, a Markdown folder

12 realistic development tasks (a landing page, an MCP server, a Claude Code plugin, a flaky API, Office files and others), each with 2 to 4 skills fixed in advance as clearly relevant, 39 in all. The same skills in three forms, 45 and 91 of them (91 is every real skill on the test machine). Every task ran once per form and size with opus through claude -p (Claude Code 2.1.310), on 2026-10-26. The prompt points at the skills, as a user would: load every relevant skill with the Skill tool; find and open every relevant skill in the KAPU file; find and read every relevant skill in this folder. A skill counts as found when the Skill tool loaded it, one of its files was read, or one of its rooms was opened.

Results per form

Wrong: loaded although the task neither needs it nor could reasonably use it. Tokens per task: every input token of the run, each turn counted (median). $ per task: what Claude Code reports at API prices, cache reads at their lower price (median). Fixed cost: the first turn of a session that needs no skill; the native skill list is in every session's system prompt.

Skills	Form	Relevant skills found	Tasks with all found	Wrong skills loaded	Tokens per task	$ per task	Calls per task	Seconds per task	Fixed cost per session (tokens)
45	Native skills	38 / 39 (97%)	11 / 12	0	42,574	$0.224	4	9	12,652
45	KAPU file, catalog (0.2.1)	38 / 39 (97%)	11 / 12	3	58,266	$0.131	5.5	25	4,775
45	KAPU file (0.2.0)	37 / 39 (95%)	10 / 12	0	77,215	$0.229	7.5	30	4,774
45	Markdown folder	39 / 39 (100%)	12 / 12	14	93,176	$0.283	10.5	24	5,430
91	Native skills	39 / 39 (100%)	12 / 12	0	44,953	$0.236	4	12	14,081
91	KAPU file, catalog (0.2.1)	38 / 39 (97%)	11 / 12	3	48,972	$0.099	5	22	4,775
91	KAPU file (0.2.0)	38 / 39 (97%)	11 / 12	0	88,418	$0.255	9	33	4,774
91	Markdown folder	39 / 39 (100%)	12 / 12	12	107,726	$0.336	10.5	33	5,430

- With a catalog of the skills in one reply (tools 0.2.1: kapu_map kind=skill, D154), the KAPU file finds as many as before (38 of 39 with 91 skills) in 5 calls instead of 9, with 48,972 input tokens per task (before: 88,418; native skills: 44,953). At $0.099 per task it costs less than the native skills ($0.236), for two reasons: it opens the first part of each skill (up to 2,000 tokens; the rest on request) where the Skill tool loads the whole skill, and less new text is written to the prompt cache, which costs twice the input price, while cache reads cost a tenth.
- The native skills still find the most (39 of 39 with 91 skills) and load nothing wrong. The KAPU file loaded 3 skills that one task did not need, and missed one skill (frontend-expert for the Figma task, which the native skills also missed with 45 skills). The Hungarian-language skill it missed before, it now finds from the catalog.
- The Markdown folder found every skill, but also read 12 skills the tasks did not need (with 91 skills) and cost the most ($0.336 per task).
- The fixed cost of a session: 4,775 tokens with the KAPU tools whatever the number of skills, against 12,652 with 45 and 14,081 with 91 native skills. 250 skills were not measured: the test machine has only 91. Only the KAPU runs were repeated; the native and Markdown runs are those of the first round.

File format 1.1: smaller files, faster calls, better search by meaning

Three changes after the pilot, measured again on the same corpora and questions. The index stores a door's position instead of its id, so files are about 4.5 times smaller and the 13-million-token MDN corpus now builds with the released tool. One call parses and checks the file once and finds index lines by binary search. Search by meaning uses one vector per 512-token window of a door, and the best window counts.

Before and after

Room recall@10: any part of the expected page among the first 10 hits. MDN with v1.0.4 was an experimental build with a 64 MiB line limit.

Metric	Corpus	v1.0.4	v1.1
File size (MiB)	Rust book	7.6	2.2
File size (MiB)	Kubernetes	117.4	26.2
File size (MiB)	MDN	542.2	128.3
Builds with the released tool	MDN	no (E005)	yes
Search over MCP (ms, median)	Rust book	56	12.8
Search over MCP (ms, median)	Kubernetes	657	19.2
Search over MCP (ms, median)	MDN	3,172	158.2
Opening a door over MCP (ms, median)	Rust book	17	2.4
Opening a door over MCP (ms, median)	Kubernetes	216	4.9
Opening a door over MCP (ms, median)	MDN	1,040	44.1
Room recall@10, hybrid search	Rust book	0.5	0.5
Room recall@10, hybrid search	Kubernetes	0.4	0.5
Room recall@10, hybrid search	MDN	0.3	0.3
Room recall@10, vectors only	Rust book	0.3	0.4
Room recall@10, vectors only	Kubernetes	0.3	0.3
Room recall@10, vectors only	MDN	0.2	0.1
kapu bench run time (s)	Kubernetes	187	8
kapu bench run time (s)	MDN	1,774	127

Same content in three formats: KAPU, one Markdown file, a PDF

The PixelLab documentation (third-party, not published), 12 requests of a beginner indie developer. The Markdown and PDF agents searched the text and read only what they found: lines of the Markdown file, pages of the PDF. Pasting the whole file into the conversation would cost 206,899 tokens on every request. Three blind Claude Sonnet judges checked every claim against the documentation.

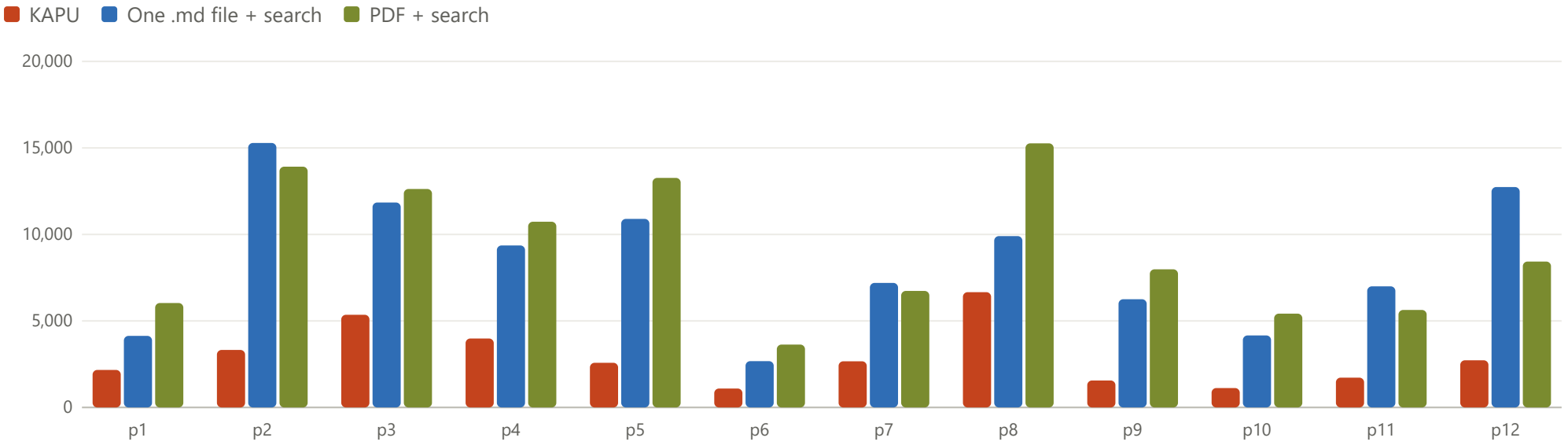
Second round: the same prompt for all three formats

KAPU with specification v1.0.4. The Markdown and PDF answers are those of the first round, scored again by new judges.

Format	Correct	Partly	Supported claims	All claims	Helpfulness (1-5)	Tokens/request	Tokens, 12 requests	Calls/request
KAPU	12	0	153	157	4.17	4,104	48,557	4
One .md file + search	12	0	140	145	4.00	8,280	101,452	8
PDF + search	11	1	147	151	3.83	8,207	109,675	6

First round: tokens read per request

KAPU with specification v1.0.3 and its own prompt. Tokens: what the tools returned to the agent (o200k_base).



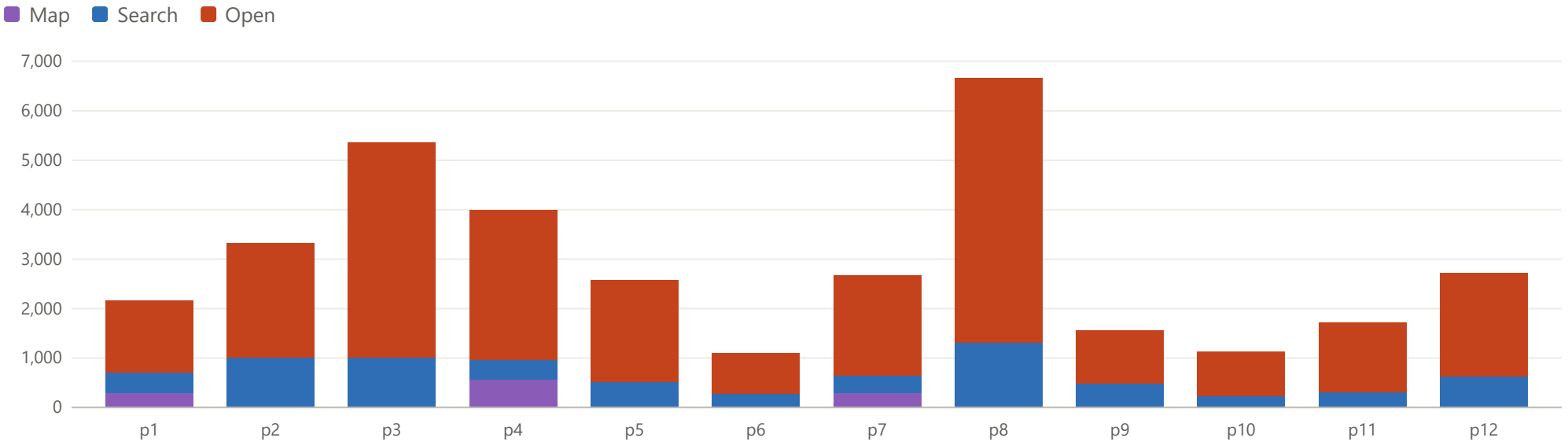
Format	Correct	Partly	Supported claims	All claims	Helpfulness (1-5)	Tokens/request	Tokens, 12 requests	Calls/request
KAPU	12	0	122	123	4.08	2,624	34,953	3
One .md file + search	12	0	134	138	4.25	8,280	101,452	8
PDF + search	11	1	146	151	4.33	8,207	109,675	6

Why thousands of tokens when the map is a few hundred?

The map and the search are the table of contents and the hit list. Most tokens come from the opened doors: the documentation text the answer is written from. Of the KAPU tokens of the 12 requests, opening doors took 77%, searching 20%, the map 3%.

KAPU tokens per request, by call

A typical request: one or two searches (300-400 tokens), one or two opened doors (1,000-3,000 tokens), rarely the map.



Scale: small, medium and large documentation

S: the Rust book, M: the Kubernetes documentation, L: MDN Web Docs, all in format 1.1. Over one MCP session: the map, every group list, 20 searches with room titles as queries and an open of each top hit.

Build, size and responses

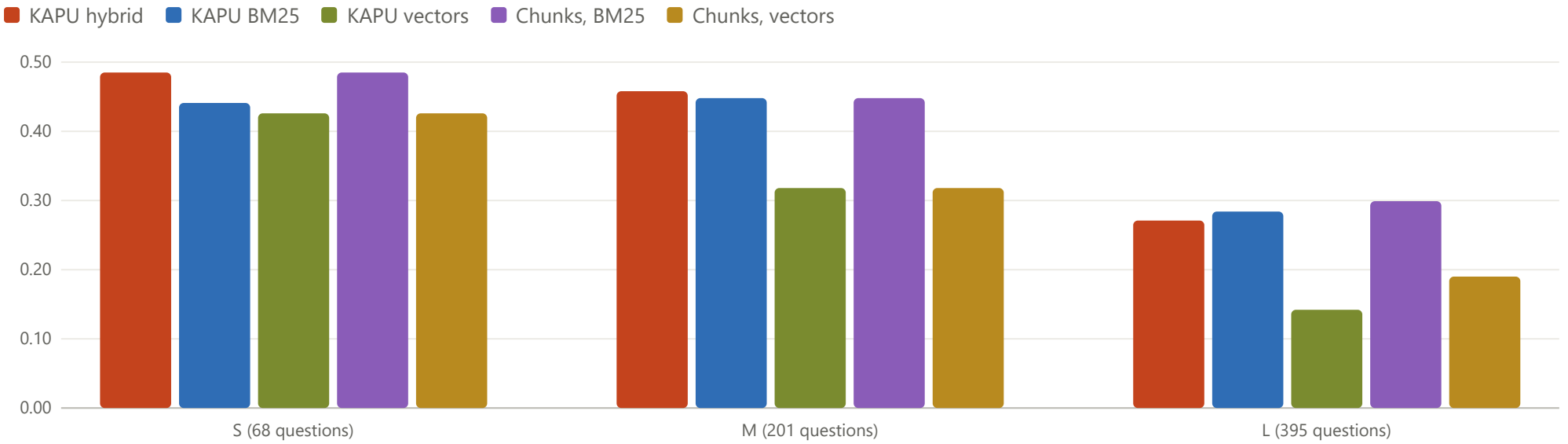
Size	Corpus	Knowledge (tokens)	Rooms	Doors	Groups	Build (s)	Memory (GB)	File (MiB)	Map (tokens)	First map (tokens)	Largest group list (tokens)	Search (tokens)	Search (ms)	First search (s)	Open (tokens)	Open (ms)
S	Rust book	321,618	111	231	7	4.6	0.69	2.2	153	322	804	246	12.8	1.1	1,689	2.4
M	Kubernetes	3,873,333	1,722	6,370	104	38	1.03	26.2	1,950	2,318	6,957	260	19.2	1	240	4.9
L	MDN	12,995,136	14,719	84,910	99	240.3	2.68	128.3	1,921	2,303	30,118	212	158.2	1.3	366	44.1

Retrieval without an agent: KAPU against chunked RAG

Real Stack Overflow questions whose accepted answer links exactly one page of the documentation: 68, 201 and 395 questions. A hit is any part of that page among the first results. The chunk baselines use 512-token chunks with 64 tokens of overlap and the same embedding model.

Room recall@10 by size and method

KAPU returns doors (rooms and their sections), the baselines return chunks.



Size	Method	Questions	Recall@5	Recall@10	MRR@10
S	KAPU hybrid	68	0.397	0.485	0.281
S	KAPU BM25	68	0.338	0.441	0.259
S	KAPU vectors	68	0.353	0.426	0.184
S	Chunks, BM25	68	0.426	0.485	0.282
S	Chunks, vectors	68	0.368	0.426	0.230
M	KAPU hybrid	201	0.373	0.458	0.245
M	KAPU BM25	201	0.348	0.448	0.218
M	KAPU vectors	201	0.219	0.318	0.147
M	Chunks, BM25	201	0.338	0.448	0.218
M	Chunks, vectors	201	0.249	0.318	0.145
L	KAPU hybrid	395	0.172	0.271	0.128
L	KAPU BM25	395	0.220	0.284	0.135
L	KAPU vectors	395	0.086	0.142	0.055
L	Chunks, BM25	395	0.230	0.299	0.159

Size	Method	Questions	Recall@5	Recall@10	MRR@10
L	Chunks, vectors	395	0.124	0.190	0.081

Live run on the Rust book: five systems, 30 questions

The model answered these popular questions well from memory too (closed book: 30 of 30), so here accuracy does not separate the systems; the cost does. KAPU with specification v1.0.3; with v1.0.4 the KAPU agents read a median of 2,331 tokens per question at the same accuracy.

Correct answers and cost per system

Two blind judges (Claude Sonnet and Haiku). All agent tokens: the whole run divided by its questions, an estimate.

System	Correct (Sonnet judge)	Correct (Haiku judge)	Questions	Tokens read/question	Calls/question	All agent tokens/question
Closed book, no tools	30	28	30	0	0	2,706
Agentic chunked RAG	30	27	30	4,150	1	12,648
grep/read on the Markdown folder	29	28	30	1,290	2	9,343
llms.txt	30	29	30	8,738	2	11,615
KAPU	30	29	30	3,982	2	10,118

Earlier fix round: specification v1.0.4

Metric	v1.0.3	v1.0.4
KAPU on the Rust book: tokens read per question (median)	3,982	2,331
KAPU on the Rust book: correct answers (of 30)	30	30
First map call, Kubernetes (tokens)	4,989	2,318
First map call, MDN (tokens)	4,139	2,303

Method and limits

- Corpora: the Rust book (MIT or Apache-2.0), the Kubernetes documentation (CC BY 4.0) and MDN Web Docs (CC BY-SA 2.5) at pinned commits. Questions: Stack Overflow API (CC BY-SA); only their ids are stored.
- Agents: Claude Sonnet subagents that could only use logged tools. Tokens are counted with o200k_base.
- Judges: blind Claude models; the answers were shuffled under random letters, and source lines were removed.
- This is a pilot on one machine (Windows 11, release build), not a benchmark. The PixelLab file is third-party and is not published.

Repository RESULTS.md Raw numbers (JSON)